

Early Design of Hydrofoils: Foil Design JIP Overview and Methods

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Abstract

The Foil Design JIP is a joint industry research project focused on enhancing hydrodynamic knowledge for the early design phase of foiling vessels. It specifically examined the performance of isolated hydrofoils in both steady and unsteady conditions, combining forced oscillation motions and free surface effects. The ultimate objective is to develop a preliminary design tool that provides comprehensive information on steady and unsteady foil characteristics, as well as relevant data for dynamic control system design. The paper describes the methodologies (numerical, experimental and data science) adopted to perform the research and the development of the tool itself. Lessons learned in the execution (and comparison) of experimental and numerical investigations are presented.

1 Introduction: Foil Design JIP and paper objectives

With reference to a hydrofoil craft sailing in full foiling mode (or foil assisted), unsteadiness due to sailing in waves can be as relevant to the performance of hydrofoil craft as the hydrodynamics in calm water, or even more important in terms of operability. It is therefore paramount that the hydrofoil design, and eventually the related control strategies, takes into account conditions beyond the steady state (Figure 1).

In the past decades a large and accurate knowledge of the hydrofoils characteristics in steady condition as been collected (typical example is [1]). This knowledge is the first basic step to consider in the design. However the hydrofoil will always operate in conditions far from steady, in that respect dedicated researches have been limited. Unsteady flow effects can have a relevant influence on the performance of the hydrofoil. In general, the amplitude of the lift forces on an oscillating hydrofoil decreases with



respect to the quasi-static solution and a phase lag with respect to the apparent angle of attack appears.



(a)



(b)

Figure 1: Example of foiling boats in different wave conditions (Left: America's Cup Boat, Right: Foiling Pleasure Craft).

A design tool capable to predict the lift and drag for an extended range of possible applications has been developed during the 'Foil Design' Joint Industry Project. The novelty of the design tool lies in fact that steady as well as unsteady conditions are considered during the early design phase. The tool can identify optimal foil designs (or predict the performance for a given input) within a design space enough extended to serve a large variety of applications. This aspect will reflect in having a more realistic awareness of the hydrofoil performances in calm water and in waves, and more importantly to gain an extended knowledge concerning the definition of the dynamic control strategy in combination with the generation of mathematical models of the hydrofoil itself.

The paper focuses on describing the pipeline of tasks required to develop its architecture and output. The purpose is to provide a global overview of the crucial parts of the tool and their development, redirecting to existing or future additional publications for more in depth explanations. Some lessons learnt along the pipeline are also presented to point out possible future topics of research and further development.

1.1 Unsteady effects: technical background and impact on performance

In general unsteady conditions are related to the action (and mutual interaction) of the following aspects:



1. Hydrofoil oscillating motions due to the wave induced orbital motions, craft motions and control surface actions;
2. Occurrence of cavitation and ventilation;
3. Hydro-elastic behaviour of the hydrofoil.

Focusing on the first point, from the pure hydrodynamic point of view, this unsteadiness can be related to the combination of the following two situations:

- Non-uniform or unsteady motions of the lifting surface in a surrounding fluid where the velocity field is uniform; as an example an hydrofoil oscillating in calm water.
- Non-uniform or unsteady inflow to the lifting surface in steady motion ("gust" problem); examples of this are hydrofoils in waves or propellers in a spatially non-uniform wake.

Associated with an oscillating hydrofoil, there are for example continuous variations in angle of attack and submergence. Their influence cannot be properly taken into account in a quasi-steady manner by considering time averaged values of these quantities only. Two unsteady flow effects play a role: first, force components proportional to the time derivative of the lift; second, memory effect on the instantaneous forces are present. The memory effect is the influence of the time dependent wake vorticity on the instantaneous foil forces: as shown Figure 2, the downwash induced at the foil by the wake is time-dependent; the strength and position of the vorticity in the vortex wake reflect the history of the motions of, and forces acting on, a foil.

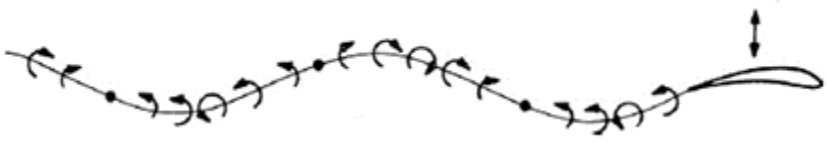


Figure 2: *Unsteady vortex wake behind an oscillating foil. The case shown is for heaving motion of the foil, with a reduced frequency of $k = 1.0$. The orientation and length of each arrow denotes the sign and strength of the vorticity at the corresponding position in the wake.* [2]

The reduced frequency "k" (see definition in the table of Appendix I) is generally used to distinguish among the following flow regimes:

- Steady for $k = 0$



- Quasi steady for $0 \leq k \leq 0.05$
- Unsteady for $k > 0.05$ ($k > 0.2$ is considered highly unsteady)

These unsteady flow effects can have a relevant influence on the performance of the hydrofoil: the amplitude of the lift forces (in general, the slope of the lift curve) on an oscillating hydrofoil decreases with respect to the quasi-static solution and a phase lag with respect to the apparent angle of attack appears. These effects increase with increasing oscillation frequency.

In the Figure 3a, 3b and 3c, experimental measurements, carried out during the JIP, of the lift coefficient of the horizontal wing (T-foil configuration) are presented as an example of the impact of the unsteady effects with respect to quasi-steady conditions. For all the compared cases, the tested speed on model scale was 4 m/s, which would correspond to 25 knots on full scale in case of scale factor 10. This assumed scale factor is considered in the rest of the paper when mentioning full scale values. More details on the tested hydrofoil (named "Foil II") and for the testing setup can be found in Appendix II. In particular, the following conditions are compared:

- Different foil immersions (h/c = 0.5 and 1.5; where "h/c" is defined in Appendix I).
- Different values of reduced frequency for forced sinusoidal oscillations in pitch of the horizontal wing (around the point 1/4 of chord from L.E.) obtained by varying the period of oscillations (on model scale between 0.36s and 1.33s, which on full scale would be between 1.15s and 4.2s).
- Different values of reduced frequency in case of incidence with head regular sinusoidal waves; in this case, the geometric angle of attack was kept constant at zero degrees and with the foil statically immersed at $h/c = 1.5$; in particular on full scale, the tested regular waves were featured by periods of about between 3 and 4 seconds and height of 1 meter. The effective variation in angle of attack was derived by considering the advance speed and the time dependent values of the velocity components of the wave orbitals at the location of the leading edge.

In Figure 3a and Figure 3b, the effects of unsteady conditions due to pitch oscillations and the effect of proximity to the free surface are presented: in general, a shallow immersion of the foil leads to a decrease of the lift curve slope; progressively high values of reduced frequency cause memory effects to become more evident with wider hysteresis loops. More in details, the following quantitative considerations could be taken for the specific foil under investigation:

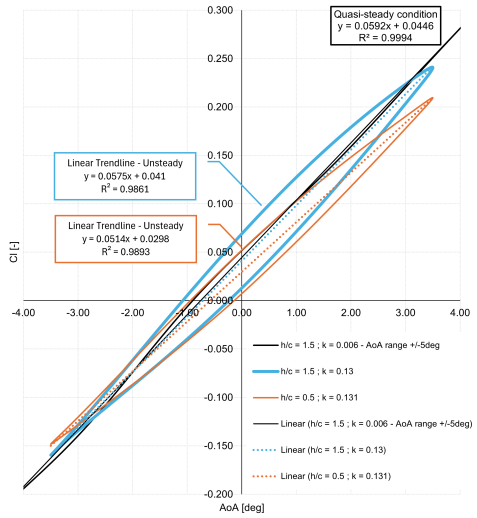
- e.g., for an angle of attack of 1 degrees the change in curve slope leads to a reduction in lift coefficient of about 5% and 22% at k = 0.13, respectively for h/c = 1.5 and 0.5, when compared to the quasi-steady condition. Similar order of magnitude in reduction of the curve slope is observed

at higher reduced frequency ($k = 0.21$), where the reduction becomes about 3% and 21% for the same values of h/c.

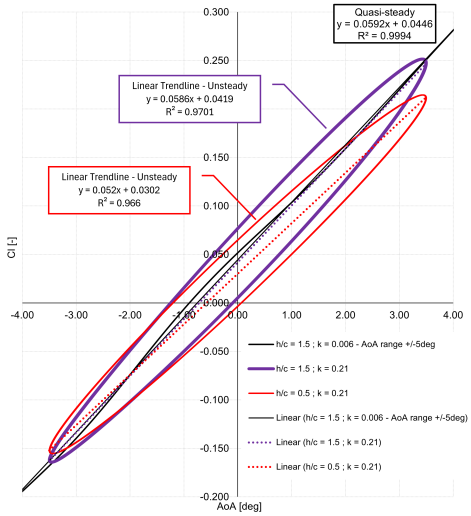
- With respect to the memory effect, the hysteresis loop leads to significant variation of the lift coefficient across the range of angle of attacks. E.g. for both considered h/c values at k = 0.13, the lift coefficient can assume a value of about -26% or +28% at Angle of Attack (AoA) of one degree with respect to the one based on a linear trend line (equation shown in the figures); at higher reduced frequency ($k = 0.21$), these variations are increased in magnitude up to -38% and +38%.

The aforementioned hysteresis loop and variability in lift coefficient becomes even more evident in the case of the hydrofoil advancing in waves: in Figure 3c, this effect is pointed out for two "extreme" tested regular head waves, which on full scale would correspond to a wave height of 1m and period of 4.4s (for $k = 0.15$) and 3.6s (for $k = 0.21$). Taking as an example the lift coefficient at zero degrees angle of attack and foil immersion of $h/c = 1.5$: the change can be of about +224% and -199% with respect to the quasi-steady condition. Even in case of advancing in waves with a reduced frequency of 0.06 (therefore almost quasi-steady conditions), variations of about +131% and -119% in lift coefficient can occur due to hysteresis effects at zero degrees AoA.

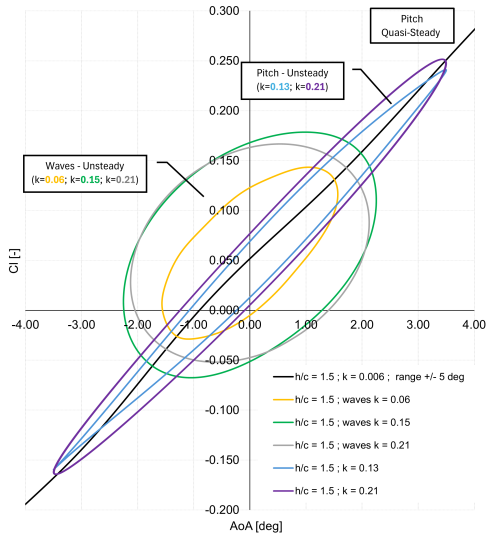




(a) *Unsteady conditions for $k = 0.13$ at two different foil immersions (respectively $h/c = 0.5$, orange line, and $h/c = 1.5$, blue line).*



(b) *Unsteady condition for $k = 0.21$ at two different foil immersions (respectively $h/c = 0.5$, red line, and $h/c = 1.5$, purple line).*



(c) *Different regular head waves, featured by three wave frequencies for $k_w = 0.06$; yellow, $k_w = 0.15$; green, $k_w = 0.21$; grey line. The lift coefficient is plotted in function of the effective angle of attack (AoA).*

Figure 3: *Foil II - Experimental measurements of lift coefficient in case of horizontal wing advancing in calm water while pitching in quasi-steady condition ($k = 0.006$) and unsteady conditions for pitch and wave conditions, respectively.*



In the literature, the work of [4] on oscillating 2D airfoils in unbounded flow (without the presence of a free surface) is one of the first references to consider the influence of unsteady effects on the lift curve slope and phase lag. Figure 4 presents the real and imaginary parts, F and G respectively, of the Theodorsen function as a function of the reduced frequency. Unsteady flow effects reduce the lift curve slope by a factor of $(F^2+G^2)^{0.5}$ and introduce a phase lag of $\arctan(-G/F)$ relative to the incidence at the 3/4 chord point from the trailing edge (TE), compared to the quasi-steady lift, which is proportional to the instantaneous incidence and a lift curve slope of 2π (according to thin airfoil theory).

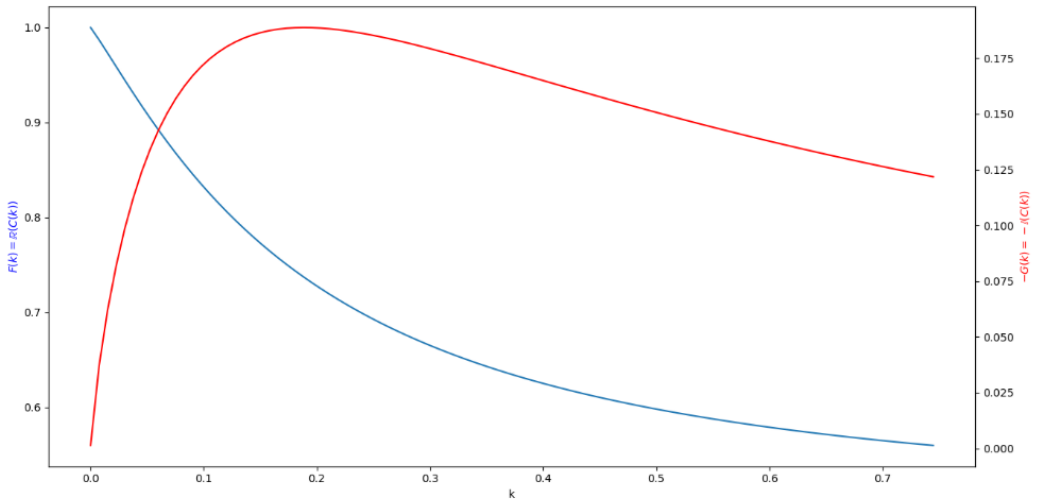


Figure 4: *Real F and imaginary G parts of Theodorsen function.*

However, the Theodorsen function does not take into account aspects such as the influence of the free surface or the three dimensional effects of finite wings. From Figure 3a and Figure 3b related to the foil immersion 1.5x chord, the curve lift slopes for forced pitch oscillations of the horizontal wing were compared with the predicted ones based on Theodorsen's function correction. This comparison is summarised in the Table 1, where, for the tested hydrofoil, the correction via Thedorsen function appears to be more conservative.

In conclusion, the effects of unsteady conditions on the performance of hydrofoils can be remarkable and can play a major role when determining the performance of the hydrofoils in realistic operational conditions and when defining the strategy of the dynamic control. A theoretical method, such as the Theodorsen function, appears to be helpful but it is based on simplifications. These effects can be tackled in advanced phases of projects for a specific design but at the cost of dedicated investigations



(numerical and/or experimental).

Table 1: *Comparison of lift slope curve values (measured and computed by means Theodorsen function).*

Condition	**Reduced frequency**	**Lift curve slope Theodorsen correction**	**Lift slope** $\frac{dC_L}{dAoA}$	
			Theodorsen correction factor	Measured
	[-]	[-]	[1/deg]	
Quasi-steady	0.006	-	-	0.0592
Forced pitch oscillation	0.13	0.82	0.0483	0.0575
	0.21	0.74	0.0441	0.0586

2 Methodology: Tool Architecture Overview

An overview of the workflow and the different building blocks forming the base of the tool's architecture is shown in Figure 5. The general idea is to use typical early design phase inputs where only limited amount of information are available to easily assist the exploration of design possibilities.

The core of the tool itself is a data driven prediction model of lift and drag generated by the horizontal wing of a T-foil configuration, excluding the influence of the vertical strut. For the latter, a preliminary estimate of its influence will be given for a range of possible strut thicknesses as a post-processing step (this feature was under development at the time of writing the current paper). The choice of a data-driven model avoids the limitations with respect to the need to run directly CFD calculations and/or to work within limited design spaces and conditions. The goal is to perform preliminary predictions and designs for a wide range of applications, representative of the market's needs: from light displacement (such as about 25 tons or less) up to medium displacement ships (such as about 70 tons or more), full foiling or foil assisted condition, wide range of speed (spacing from 7 up to 40 knots). The data-driven model allows to predict lift and drag in quasi-steady and unsteady conditions for three situations: pure pitch and heave sinusoidal motions of the wing and advance through regular incident waves. The pitch oscillations are performed around the rotation point placed at 1/4 of the chord from the leading edge; the heave oscillations are considered by keeping the hydrofoil at the geometrical zero degrees angle of attack; the latter is also kept when advancing in incident waves (only head waves). For the cass with imposed motions or with the incident waves, the oscillation frequency can be varied in order to reach



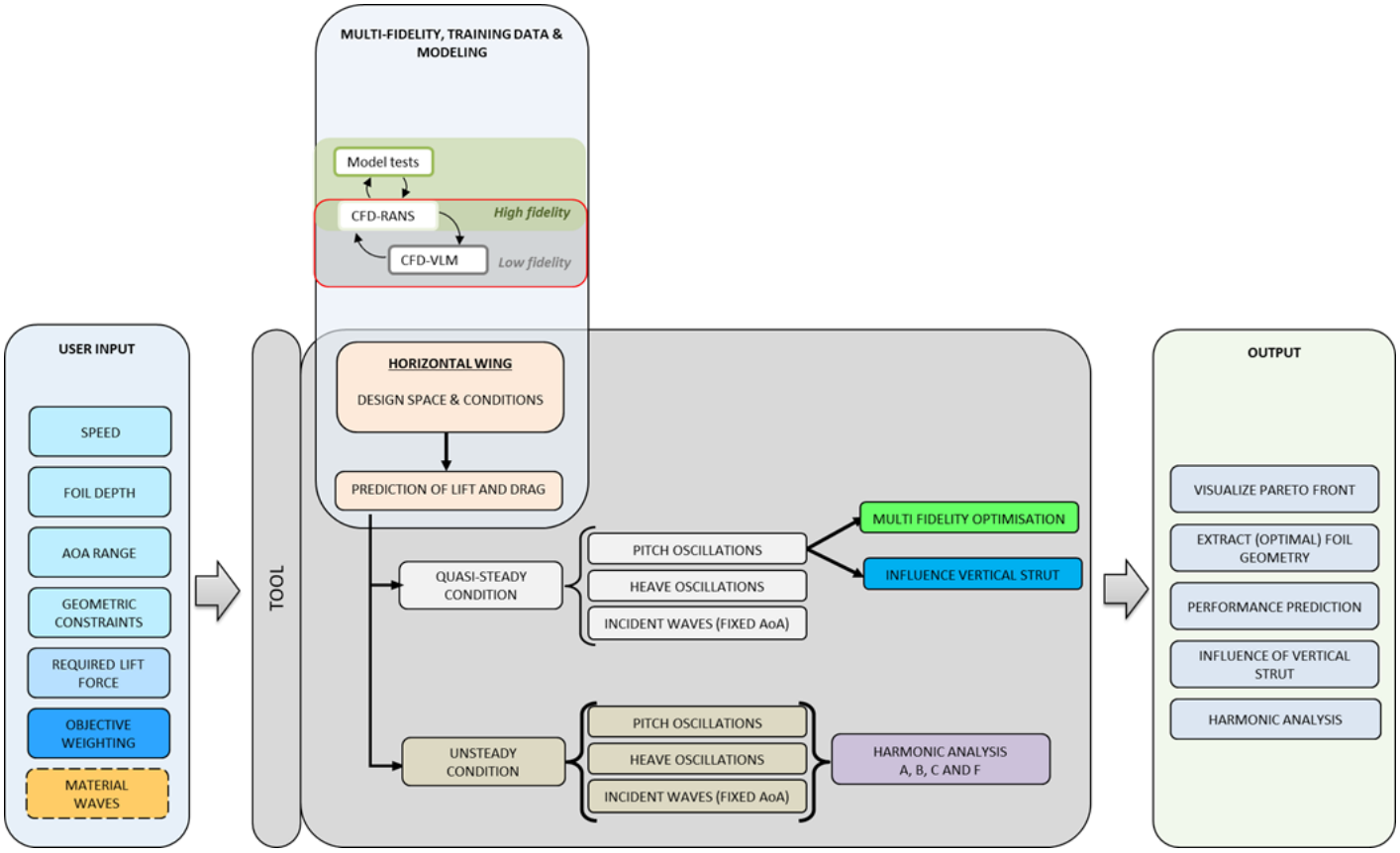


Figure 5: *Foil Design JIP tool architecture and workflow*

different levels of reduced frequency in combination with the given speed. The usage of the tool goes through two possible linked modes: optimisation or evaluation. In the first case, it is possible to identify an ensemble of possible optimal designs (Pareto front), while in the second case the lift and drag are predicted for a selected design, whose geometrical features and operational conditions are specified by the user. An interactive link is possible between the two modes: once a hydrofoil is selected from the Pareto front, further investigations about its performance (at different speeds, depths, reduced frequency values and in incident waves) can be carried out in the evaluation.

Regarding the optimisation mode, a time averaged prediction is used to compute the performance of different generations of hydrofoils, which are progressively created by the underlying genetic algorithm. Ultimate result is a Pareto front, from which the user can select optimal design options. In addition to time-averaged (quasi-steady) predictions, unsteady condition calculations allow for performance evaluation early in the design phase without requiring dedicated investigations. These calculations also provide



the necessary coefficients for the equation of motion (see Appendix I), useful to set-up dynamic control strategies and mathematical models of the hydrofoil behaviour in simulator environments.

Based on the previously described aspects, the output of the tool collects: interactive visualisation of the Pareto front, extraction of the hydrofoil geometry (planform and wing profile), prediction of lift and drag in function of angle of attack and time (for unsteady conditions), estimate of the vertical strut influence on the lift generated by the horizontal wing in quasi-steady conditions and, last but least, harmonic analysis to derive coefficient of equation of motions. The development of the data-driven model, as also indicated in Figure 5, allows the prediction of the lift and drag without the necessity to run direct calculations. Foundations and accuracy of this model were established by performing different activities:

- Experimental model test results were considered as true values in validation of the CFD-URaNS calculations (taken as high fidelity);
- Large quantity of calculations performed by means of CFD-URaNS (Unsteady Reynolds averaged Navier-Stokes equations) and CFD-VLM (Vortex Lattice Method), respectively adopted as high and low fidelity calculations;
- Data science methodology, described on a more general level in the current paper while more details can be found in [3];
- Hydrofoil design strategy in quasi-steady condition (evaluation of cavitation buckets, minimum thickness for yield strength and vertical strut influence).

The experimental results were obtained by model tests in the Technical University of Delft (TU Delft), which joined the JIP with this activity as in-kind contribution.

2.1 Data-driven modelling

In terms of accuracy, the performance of the prediction model is strictly linked to the quality and quantity of the training data. Therefore, the challenge to identify enough large datasets, especially when high-fidelity simulations like CFD-URaNS are computationally expensive and time-consuming. In that respect, instead of relying solely on high-fidelity data, a multi-fidelity surrogate modeling approach was adopted for the development of the prediction model at the core of the tool: large datasets of low-fidelity simulations, such as those from the CFD Vortex-Lattice Method, were combined with a smaller number of high-fidelity CFD-URaNS calculations.

Fundamental to this approach is a bi-directional Long Short-Term Memory (LSTM) model. The LSTM model is particularly suited for capturing complex, time-dependent relationships in sequential data, enabling accurate prediction of dynamic behaviors such as lift and drag. By processing information from both past and future time steps, the bi-directional LSTM effectively models the interactions between pitch, heave, and wave influences, resulting in greater predictive accuracy for these critical performance metrics. The general modelling workflow is presented in Figure 6, however for further technical details of the surrogate modelling and training methodology, refer to [3].

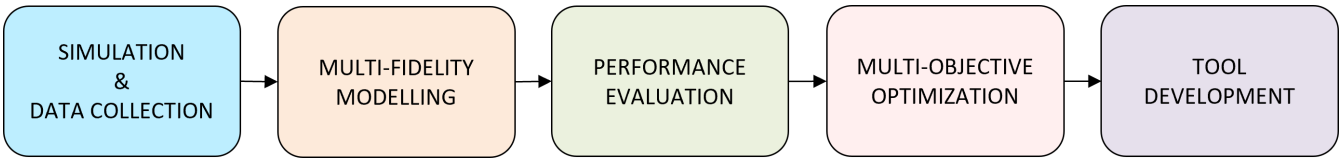


Figure 6: General modelling workflow

The surrogate models created through this multi-fidelity framework act as simplified representations of the more complex, high-fidelity simulations. This allows for rapid exploration of the design space while maintaining sufficient accuracy, especially in the early stages of design when numerous evaluations are necessary. Low-fidelity models provide broad approximations, and high-fidelity models are selectively employed to refine the most promising designs.

By integrating these different fidelity levels, the framework enables faster convergence toward optimal solutions while ensuring that the final designs are feasible, high-performing, and computationally efficient. For future evolution of the tool, as we continue to generate more data from both low - and high-fidelity sources, the predictive power of the model will only increase, leading to more reliable and accurate design outcomes.

2.2 Design space and conditions

To explore the most representative design space, a dimensional analysis was conducted using the Buckingham Pi theorem. This method was used to derive a set of core non-dimensional parameters (Π-groups) that fully describe the physical phenomena affecting the hydrofoil design. These Π-groups were then used non-dimensionalise the problem, allowing for more efficient exploration of the design space.

Non-dimensionalisation offers significant advantages, particularly when sampling the design space. It ensures that all solutions respect the physical constraints governed by natural laws. To sample this design space while balancing the computational cost with performance accuracy, a nested Latin Hypercube Sampling (LHS) approach was employed. This nested LHS method allowed for strategic and



cost-effective coverage across the identified 11 key non-dimensional parameters, which included the Froude number (Fn), Reynolds number (Re), Aspect ratio (AR), and others like submergence ratio (λ_s), and Strouhal number (St), the dimensional analysis effectively captures the essential geometrical and operational relationships.

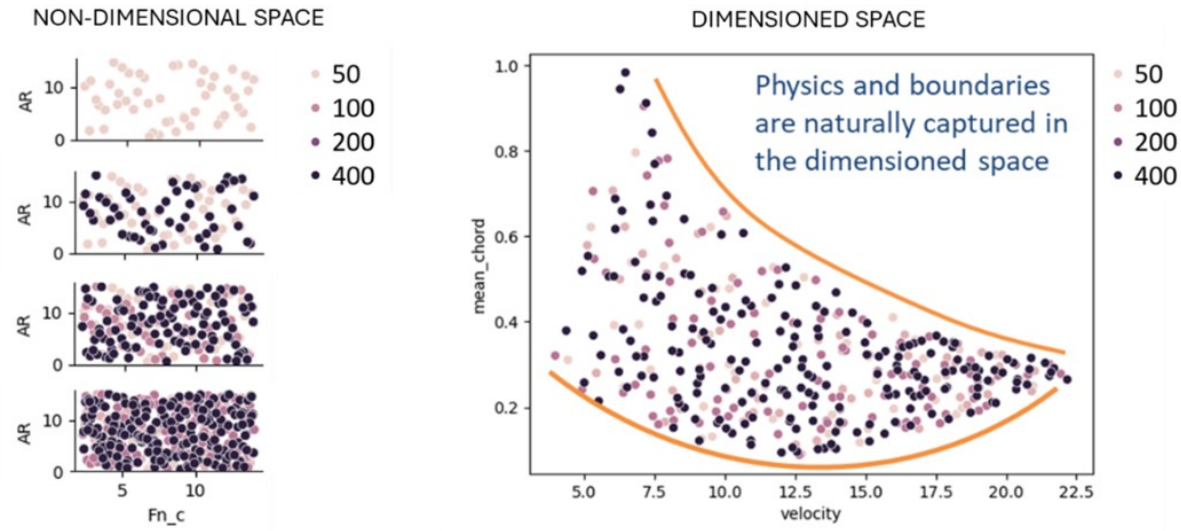


Figure 7: *Example of correlation between non-dimensional and dimensional space.*

This approach not only ensures that the design exploration covers plausible, physically meaningful regions of the parameter space, but also allows for an easier transition back into fully-dimensioned space; furthermore, it helps to preserve the integrity of the physical models while ensuring broad coverage of the design space for data-driven approaches. Figure 7, helps to demonstrate this effect. As we sample more points in the non-dimensional space, in this case considering the aspect ratio of the foil with respect to the Froude number, we can see that the corresponding dimensioned samples fall within a constrained physical space. Anything beyond these boundaries are physically not a suitable solution subject to the inferred modelling assumptions and data ranges. As a result, the designs sampled are not only theoretically valid but also practical in real-world applications. For further technical details on the dimensional analysis and the specific Pi groups derived, refer to [3]. A detailed description of the full design space in both dimensional and non-dimensional space along with their extents is summarised in Table 2, while a sample of the possible foils shapes considered within the tool's design space is presented in Figure 8.



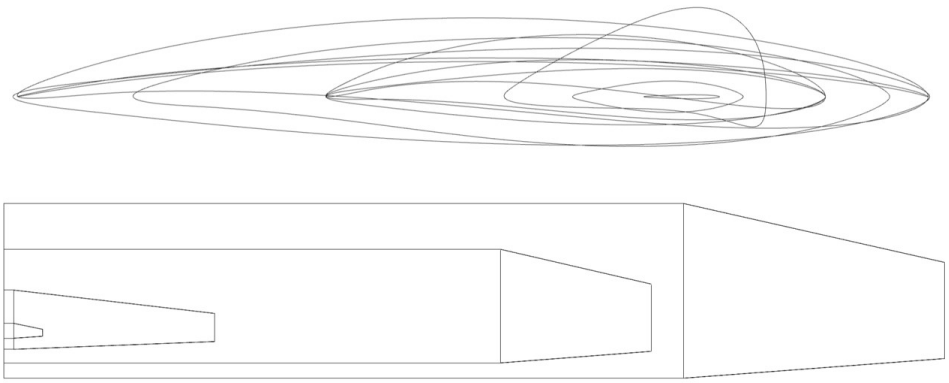


Figure 8: *Sample of the possible foils shapes (wing profiles and planforms) within the tool's design space.*

2.3 Multi-objective optimisation

The hydrofoil design optimisation process employs a multi-objective surrogate-based optimisation approach, aiming to balance several key performance objectives while accounting for complex constraints. The optimisation uses NSGA-III (Non-dominated Sorting Genetic Algorithm III), which is well-suited for handling multi-objective, non-linear, and multi-modal problems. NSGA-III ensures diversity along the Pareto front, a set of non-dominated solutions representing trade-offs between competing objectives (e.g., improving lift may increase drag).

The formulation of the optimisation problem involves minimising a set of performance functions while satisfying structural and operational constraints. For further technical details, refer to [3]. The data-driven surrogate models are used to approximate these objectives and constraints, which significantly reduces computational cost compared to running full simulations. The key objectives are:

1. Maximise Mean Lift: This is a critical metric for hydrofoil design, as it determines the foil's ability to support weight and generate vertical force. Mean lift is calculated using a quasi-steady approximation model.

2. Minimise Mean Drag: Reducing drag is essential to increase the efficiency of the hydrofoil, allowing for higher speeds and lower energy consumption. Similar to lift, drag is calculated using a quasi-steady model.

3. Minimise Cavitation Risk: Cavitation occurs when local pressure drops below vapor pressure, can damage the foil and degrade performance. The risk of cavitation is evaluated using cavitation



Table 2: *Complete design space feature range extents*

Non-Dimensional								
Description	Symbol	units	Heave		Pitch		Wave	
			minimum	maximum	minimum	maximum	minimum	maximum
Froude chord	Fn_c	-	2.016	13.988	2.016	13.988	2.016	13.988
Reynolds chord	Re_c	-	1010000	5990000	1010000	5990000	1010000	5990000
Aspect ratio	AR	-	0.53	14.982	0.53	14.982	0.53	14.982
Submergence ratio	λ_s	-	1.008	4.995	1.008	4.995	1.008	4.995
Thickness ratio	λ_{th}	-	0.05	0.3	0.05	0.3	0.05	0.3
Camber ratio	λcr	-	0	0.05	0	0.05	0	0.05
Tapering ratio	λ_{tap}	-	0.501	0.8	0.501	0.8	0.501	0.8
Taper distance ratio	λ_{yk}	-	0.301	0.799	0.301	0.799	0.301	0.799
Reduced frequency	k	-	0.011	0.4	0.011	0.4	0.001	0.01
Strouhal number	St	-	0	0.003	-	-	-	-
Angle of attack	α	-	-	-	0.021	9.99	-	-
Wave height/ Depth ratio	λ_{wa}	-	-	-	-	-	0.1	0.2
Dimensioned								
Description	Symbol	units	Heave		Pitch		Wave	
			minimum	maximum	minimum	maximum	minimum	maximum
Mean chord	c_{mean}	m	0.089	0.983	0.089	0.983	0.089	0.983
Root chord	c_{root}	m	0.096	1.181	0.096	1.181	0.096	1.181
Tip chord	c_{tip}	m	0.047	0.651	0.047	0.651	0.047	0.651
Span	b	m	0.082	12.738	0.082	12.738	0.082	12.738
Projected area	A	m2	0.007	11.616	0.007	11.616	0.007	11.616
Velocity	V	m/s	3.959	22.177	3.959	22.177	3.959	22.177
Depth	d	m	0.16	4.169	0.16	4.169	0.16	4.169
Distance to tapering	yk	m	0.037	9.204	0.037	9.204	0.037	9.204
Camber	cam	m	0	0.047	0	0.047	0	0.047
Thickness	th	m	0.005	0.245	0.005	0.245	0.005	0.245
Frequency	f	rad/s	0.05	16.565	0.05	16.565	0.003	0.417
Heave height (peak-peak)	A_{heave}	m	0	0.423	-	-	-	-
Angle of attack	α	deg	-	-	0.021	9.99	-	-
Wave height (peak-peak)	A_{wave}	m	-	-	-	-	0.018	0.745
Froude depth	Fn_d	-	0.619	17.687	0.619	17.687	0.619	17.687
Constraints								
Avoid foil emergence	Heave constraint	$(\pi \cdot St)/k < \lambda_s$						
Avoid non-physical wave Steepness	Wave constraint	$A_{wave}/(T_p^2 \cdot g/(2\pi)) > 1$						
Avoid foil emergence in waves	Wave cons. 2	$2/\lambda_{wa} > 1$						

buckets, which are computed for different angles of attack and operating conditions.

In addition to the objectives the optimiser is also constrained, such that the produced foil geometries

are both technically and practically viable solutions. The following constraints are considered:

1. Minimum Required Lift: The foil must generate sufficient lift to support operational loads. As such, a minimum threshold must be satisfied to be useful. Optionally, and upper limit constraint can also be applied in which an exact target lift must be obtained.

2. Bending Stress Constraints: The foil's structure must remain within the material's yield limits to prevent failure. This is evaluated using the bending stress, calculated based on the cross-sectional properties and moments acting on the foil.

3. Zero Neutral Angle Cavitation: To avoid cavitation under typical operating conditions, the design must ensure that cavitation does not occur at a neutral angle of attack. This is evaluated using the cavitation bucket approach to ensure the evaluated hydrofoil does not exhibit any neutral angle cavitation risk.

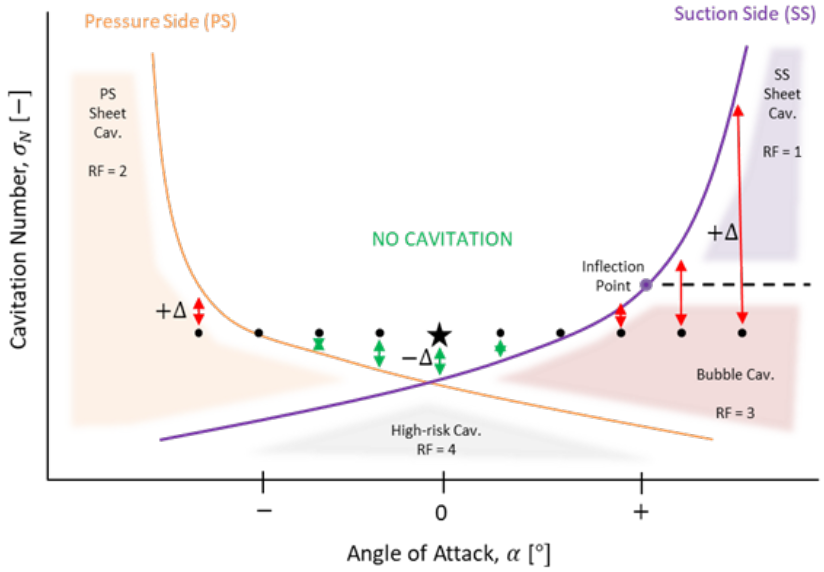


Figure 9: *Cavitation risk objective schematic.*

Figure 9 illustrates a generalised cavitation bucket for a singular foil, where each region of interest is highlight including the associated cavitation zones. Cavitation risk is quantified by evaluating cavitation buckets for sampled foils using a k-nearest neighbor (k-NN) algorithm. This approach allows the approximation of cavitation behavior for new hydrofoil designs by referencing similar designs from a pre-existing database of 400 numerically evaluated cavitation buckets (or maps). The objective function



for cavitation risk sums the differences between the predicted and threshold cavitation numbers on both the pressure and suction sides of the foil, with different risk factors (RF) assigned to different types of cavitation (e.g., bubble cavitation is the most severe).

Upon evaluation from the optimiser, the users can select the most appropriate design by applying a weighted sum of the objectives (lift, drag, cavitation risk) according to their priorities. In summary, this optimisation process allows for efficient exploration and evaluation of hydrofoil designs, balancing lift, drag, and cavitation risk while ensuring structural integrity. The use of surrogate models and genetic algorithms enables a robust and computationally feasible approach to hydrofoil design optimisation.

2.4 General modelling performance and accuracy

The overall accuracy of our data-driven model was evaluated using standard data modelling pipelines and error metrics such as the Mean Relative Error (MRE) metric. For a comprehensive discussion of the metrics, error analysis, and the detailed validation process, the reader is referred to the accompanying technical paper [3].

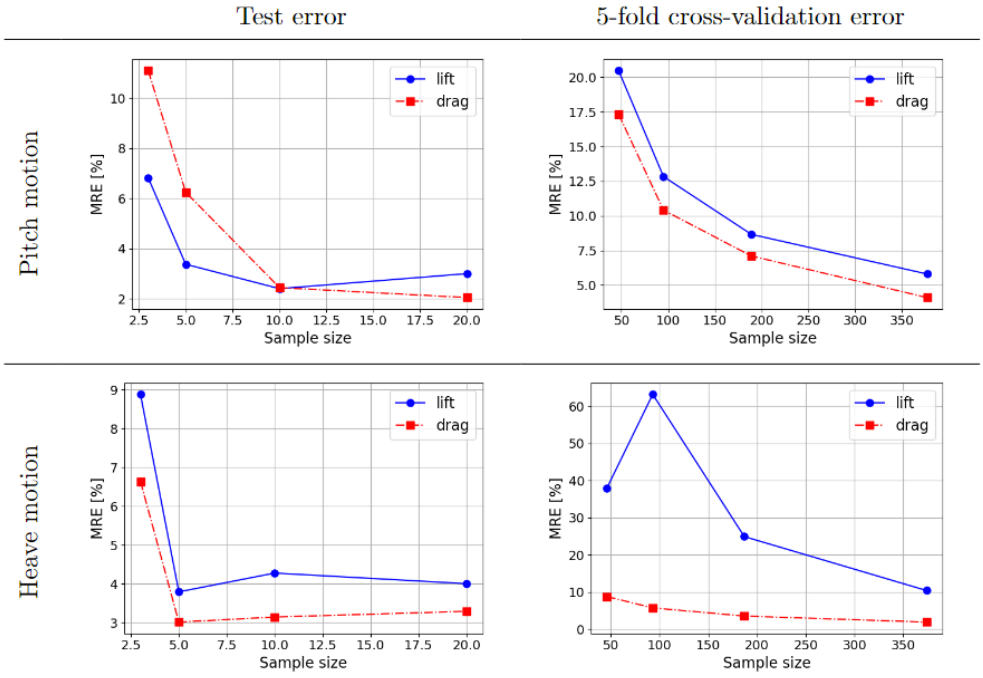


Figure 10: *Model accuracy based on validation data, with the predicted values for the forced pitch and heave models.*



As shown in Figure 10, the model demonstrates a high level of accuracy across the validation dataset, with predicted values closely matching the test data (2% - 12% error). The presented results currently, only reflects the low-fidelity model approximation as the full multi-fidelity pipelines was still under-development at the time of writing. Nevertheless, such performance provides a high degree confidence in relation to the high accuracy outcomes of the full multi-fidelity model.

3 Example Case Application

At the time of writing the current paper, the tool was still under development. Some implementations and updates were not yet available such as an update of the data-driven model based on the combination of low and high fidelity data (the tool version here considered was still based on only low fidelity data from CFD-VLM calculations). Other components to be added were the prediction of the influence of the waves on the advancing hydrofoil and the preliminary estimation of the vertical strut influence. However, an example of application is here presented to show the outcome of the basic functionalities previously described. Furthermore, it is possible to see some of the aforementioned physical trends (effect of foil immersion and increased reduced frequency).

Table 3: *Tool application example: overview of inputs for optimise mode*

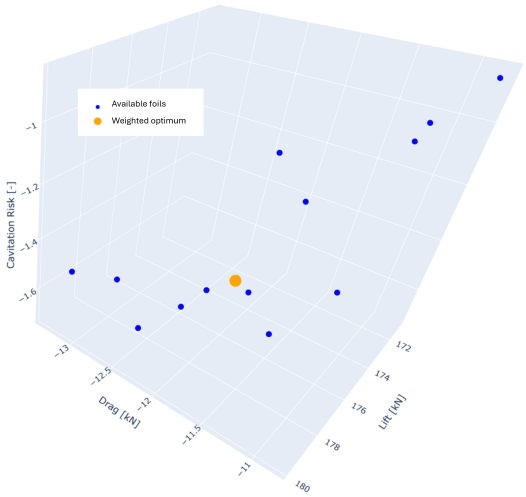
Operational inputs				
Speed	[kn]	30		
Foil depth below free surface	[m]	1.15		
Maximum angle of attack range	[deg]	5		
Heave height	[m]	0		
Required lift	[kN]	170		
Maximum bounded lift	[kN]	180		
Foil geometrical boundaries				
		Set min	Set max	Max range
Chord root	[m]	0.4	1	[0.1 – 1.181]
Tapering ratio	[-]	0.5	0.8	[0.5 – 0.8]
Span	[m]	0.1	5.144	[0.1 – 12.7]
Tapering distance ratio	[-]	0.3	0.8	[0.3 - 0.8]
Thickness ratio	[-]	0.05	0.15	[0.05 – 0.3]
Camber ratio	[-]	0.001	0.05	[0.001 – 0.05]
Objective importance weighting				
Lift	[-]	0.33		
Drag	[-]	0.33		
Cavitation	[-]	0.33		



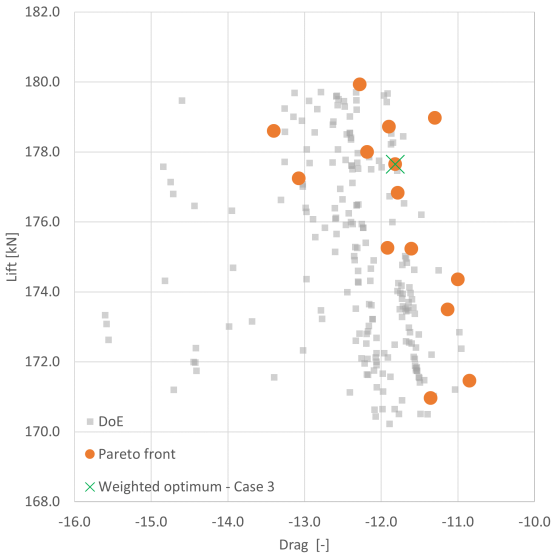
First task of the example is to generate a set of preliminary foil designs delivering the target lift at a chosen design speed and within certain limits of main dimensions. The summary of the inputs is given in Table 3.

Part of the interactive outcome of the tool's user interface is shown in Figure 11a, where only the optimal designs on the Pareto front are visualised. Given the three objective functions (reach target lift, minimise drag and minimise the cavitation risk), the points of the Pareto are distributed in a three dimensional space: a two dimensional visualisation is also possible as shown in the three remaining images of Figure 11, where not only the Pareto front but also the underlying automatically generated Design of Experiments (DoE) are presented in function of the combination of the three aforementioned objective function. One specific design (Case "3") is pointed out, being the resulting optimum for the selected objective importance weighting.

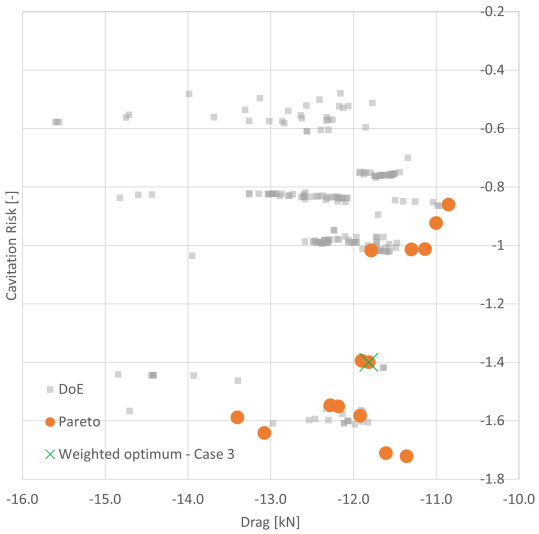




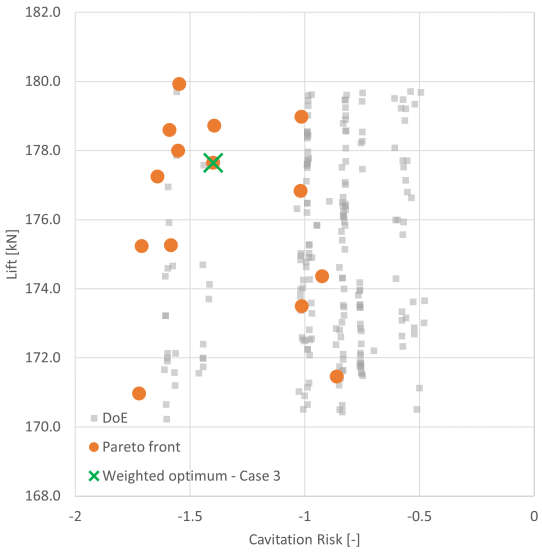
(a)



(b)



(c)



(d)

Figure 11: *Example of application - (11a) Three dimensional overview of Pareto front from tool's user interface. (11b, 11c, 11d) Two dimensional visualisation of the tool's automatically considered design of experiments (DoE in grey dots) and resulting Pareto front (orange dots); from the Pareto front one case is pointed out (green cross) as the optimum based on the selected objective importance weighting.*



Figure 12 gives an comparison overview of the resulting planform shapes and wing profiles for the aforementioned Pareto front. The ultimate decision of which design take from the Pareto front is case and user dependent.

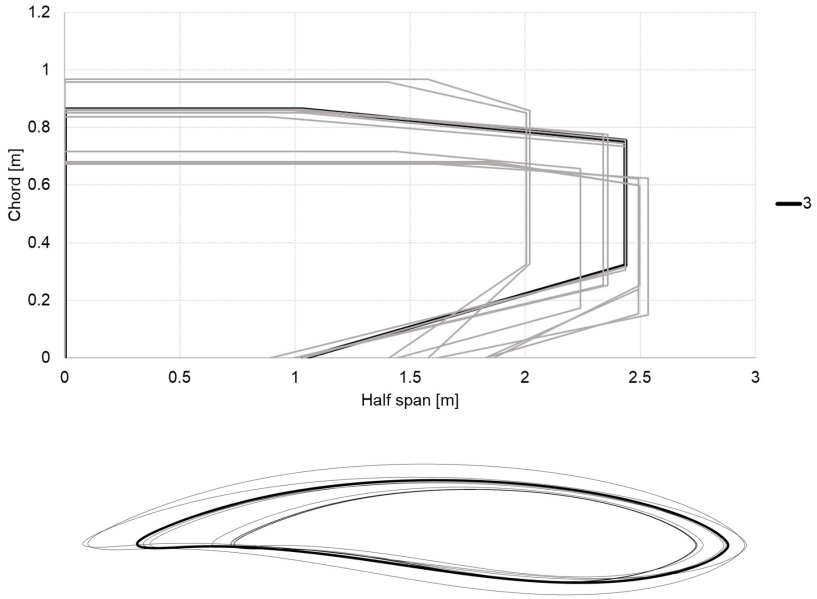
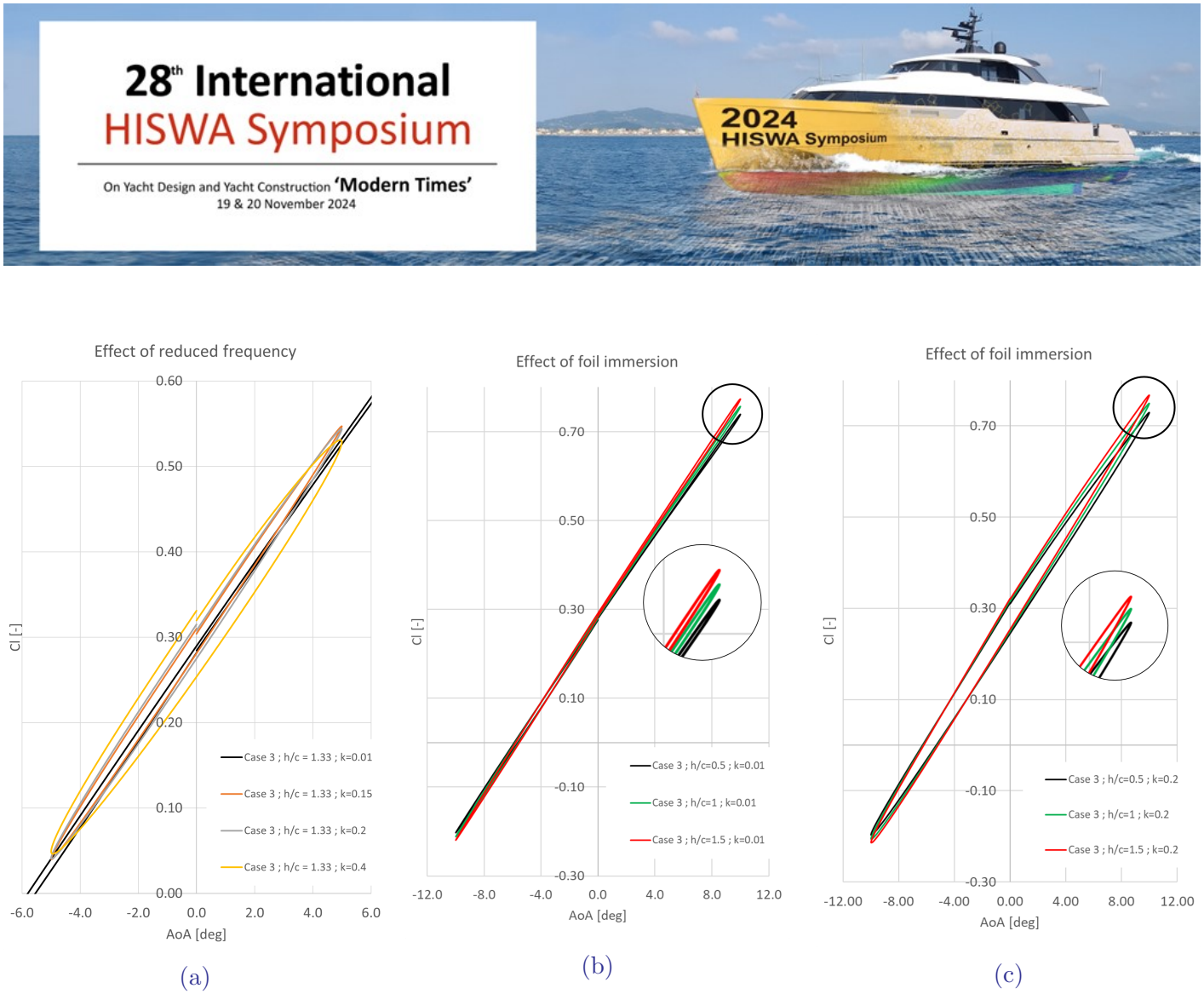


Figure 12: *Example application - Comparison of planform shapes and wing profiles from the Pareto front (pointed out is Case 3 in solid black line).*

Once identified an optimal starting point (Case 3 in this specific example), additional analysis can be carried out in the Evaluate mode of the tool to further explore the performance of the selected foil under conditions beyond the design point: different foil immersions, larger ranges of angle of attack, increased values of reduced frequency for forced pitch or heave oscillation, influence of encountering different type of head waves. In Figure 13 this possibility to explore different operative conditions is presented such as: the influence of progressively increased unsteady conditions (increasing reduced frequency value) or the dependency of the lift curve slope to the foil immersion. The outcome of the tool presents similar trends as shown in the introduction concerning these effects.



(a) (b) (c)

Figure 13: *Example application – Case 3 from Pareto Front; Figure 13a shows the influence of reduced frequency (progressive increased unsteady conditions) at the same foil immersion of the foil ($h/c = 1.33$); Figure 13b and Figure 13c show the influence of foil immersion ($h/c = 0.5$, 1 and 1.5) respectively for $k = 0.01$ and $k = 0.2$.*

4 Lessons Learned

Several lessons were learnt during developments of the tool during the model test campaign, the development of the data-driven model and also the execution of the CFD calculations. For sake of brevity of the paper, not all of them can be here discussed. However, some of them are here briefly mentioned.

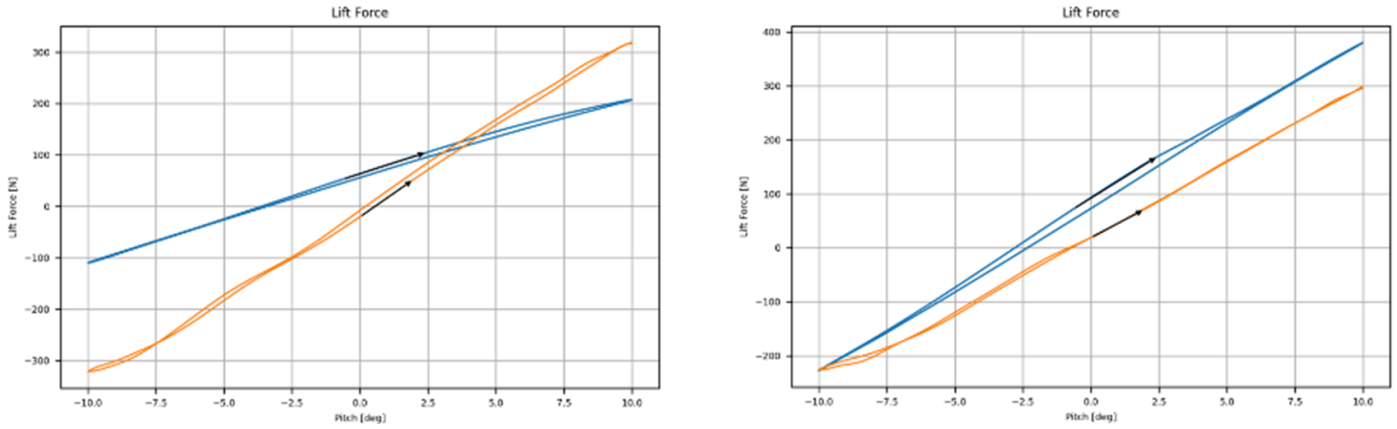
Amount (and quality) of training data: The accuracy of data-driven models are dependent on the amount (and good quality) of training data. Given the accuracy metric discussed in section 2.4, additional computations (low and high fidelity) would improve the accuracy and reliability of the prediction model as visualised by the learning plot convergence. The imposed training philosophy focused on sampling and covering the design space as broadly as possible by application of modern "space-filling"



approaches. While, this has the benefit of expanding the general modelling performance, alternative methods exist to smartly target "zones" which exhibited the largest degree of uncertainty. Given the high cost of obtaining training data, the strategy is to selectively add data points in areas where coverage is sparse, thereby increasing the model's predictive accuracy and performance.

Influence of flexibility: The model test campaign was meant to provide validation material for the CFD-URaNS calculations. Three foils were tested (see Appendix II) but for two of them (Foil I and III), the measurements could not be used because the results were affected by a larger than expected elastic behavior. In Figure 14, the measurements carried out in quasi-steady conditions for Foil I and Foil II are compared with the CFD-URaNS calculations performed at the same conditions. In the numerical calculations the foils are modeled as rigid bodies. Results show the lift curve slope between experiments and calculations was very different in case of Foil I. On the other hands, for Foil II there was a good agreement. Same behavior as Foil I was observed also for Foil III.

This incident highlights how significant the changes in hydrodynamic behavior can be when flexibility is considered and how this could potentially be leveraged for the performance and dynamic control of the hydrofoils.



(a) (b)

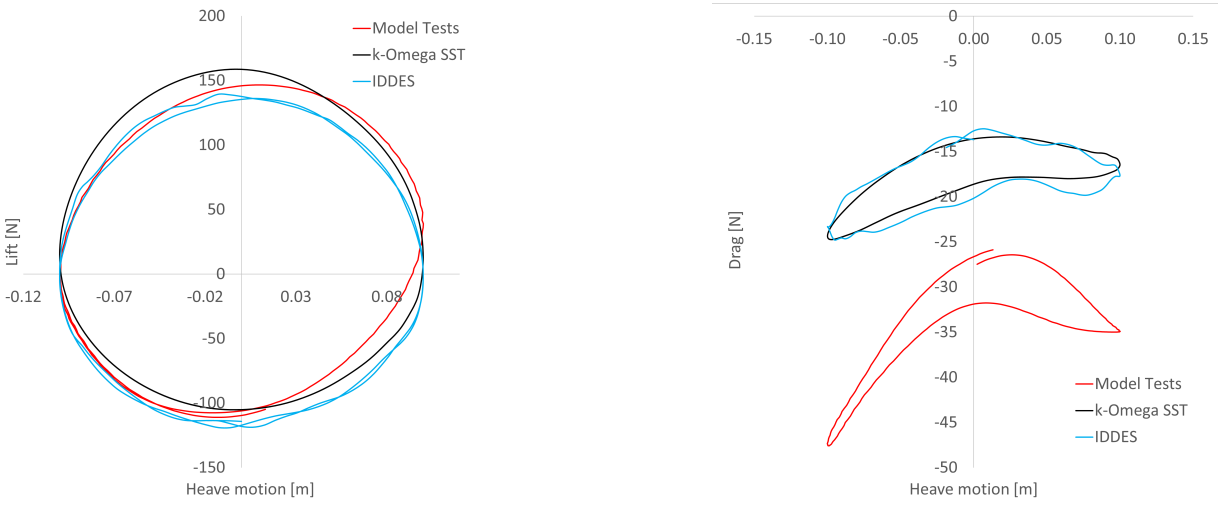
Figure 14: *Measured lift force [N] on model scale at* $4\,m/s$ *speed of advance in* $h/c = 0.5$ *immersion of Foil I (left) and Foil II (right). Orange lines are for measurements while blue are for calculations. Arrows indicate the starting point of the oscillations.*

Influence of modelled geometry in CFD: As an additional note for Foil II, the offset between the tested and computed curves seen in Figure 14 could be then reduced once the calculations were performed by removing the buoyancy force from the lift (as delivered in model tests) and, especially, by

modelling in CFD the geometry as close as possible to the tested one. In particular two aspects played a major role: the manufactured trailing edge resulted in a slightly shorter chord length with respect to the modelled geometry in CFD (due to the accuracy and production limits of the 3D printing machine) and, secondly, the connection part between the wing and the actuation system inside the vertical strut (a figure of the set-up is given in Appendix II).

Turbulence modelled in CFD: The choice of the turbulence model in CFD plays a relevant role in the lift and drag prediction. When advancing and simultaneously heaving the linear eddy-viscosity model $k-\omega$ SST over estimates the turbulence around the foil, leading to an over prediction of the lift and under prediction of the drag as shown in Figure 15.

The CFD results so far presented in the paper were all performed using the turbulence model $k-\omega$ SST. This is suitable for steady or mildly unsteady flows with limited separation; it is computationally cheap but it assumes isotropic turbulence (same turbulence properties in all directions), which limits the accuracy in complex 3D flows. In an attempt to improve the agreement between CFD and model tests, additional calculations were carried out to quantify whether different turbulence models could yeld to better results. In particular, another approach considered is to use a IDDES (Improved Delayed Detached Eddy Simulation) turbulence model, which blends from a RANS turbulence model near the wall to a LES model further away.



(a) (b)

Figure 15: *Comparison measured and computed lift and drag in forced quasi-steady heave oscillations with different applied turbulence and geometry modelling.*



From Figure 15, the IDDES led to a lift prediction in quasi steady tending towards the model test results, while the drag prediction does not show significant improvement. However, IDDES turbulence models require a larger spacial and time scale resolution than linear eddy-viscosity model such as $k-\omega$ SST. Because the same grid density and time scale was used in the comparison, better results can be expected with higher grid density and lower time step resolution. Such investigation was out of the current scope of work and it is recommended as future work.

Concluding, the adoption of more advanced turbulence models has the potential to improve the accuracy for the prediction of lift and drag but it comes at an increase of computational time, limiting the number of producible training data.

For future developments of the training set data, the application of IDDES on a large number of cases seems an promising approach to improve the accuracy of the data-driven model.

5 Conclusions

The research of the Foil Design JIP has allowed not only to widen the knowledge of the effect of unsteady conditions on the performance of hydrofoils but also to include it since early design stage. A design tool for preliminary phases has been developed to provide comprehensive information on steady and unsteady foil characteristics, as well as relevant data for dynamic control system design. A large variety of applications can be covered given the design space underlying the tool. Free surface effects and effects of unsteady conditions have been measured experimentally and their trends successfully reproduced with the prediction model at the core of the tool.

Some of the mentioned lessons learnt have set the future developments of the prediction capability of the tool but also inherent limitations due to the adopted methods:

- The work performed to develop the tool has laid the foundations and the architecture to further improve the prediction capabilities of the tool: e.g., increment of training datasets and/or dedicated techniques can be used to improve the accuracy in the areas of the design space where the tool lacks of enough data for more accurate predictions.
- The hydro-elasticity can play a fundamental role in the design and performance of the hydrofoils;
- The accuracy of CFD calculations (here treated as high fidelity data) depends on multiple factors such as proper modelling of not only the hydrofoil geometry but also of the turbulence. Regarding the latter, the application of more advanced turbulence models (such as IDDES) on a large number of cases represents a promising approach to improve the capacity of the data driven model to

predict complex conditions such as heave oscillations.

Future evolution of the described tool involves the possibility to extend it to the design and prediction of an entire foiling craft. Furthermore, its predictions could be used also at the base of simulators. The influence of ventilation, especially its inception, could be integrated in the optimisation and design process.

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Giancarlo Marelli: Methodology: Simulation, Writing: review & editing, Data curation, Validation.
Thomas Scholcz: Methodology: data analysis, Data curation, Validation.

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[1] I. H. Abbott and A. E. Von Doenhoff. *Theory of wing sections, including a summary of air-foil data*. New York: Dover Publications, 1959.

[2] J. N. Newman. *Marine Hydrodynamics*. MIT: MIT Press Academic, 2018.

[3] K. Odendaal T. Scholcz. "Provisional: Unsteady Hydrodynamic Load prediction on Hydrofoils: Development of an Early Stage Design Tool". en. In: *Pre-print* 1 (2024).

[4] T. Theodorsen. "General Theory of Aerodynamic Instability and the Mechanism of Flutter". en. In: *NACA Techical Report Number: NACA-TR-496* 1 (1949).



Appendix I

A Geometrical Parameters and Coefficients

In this appendix, we summarise the key geometrical parameters and coefficients used in this paper.

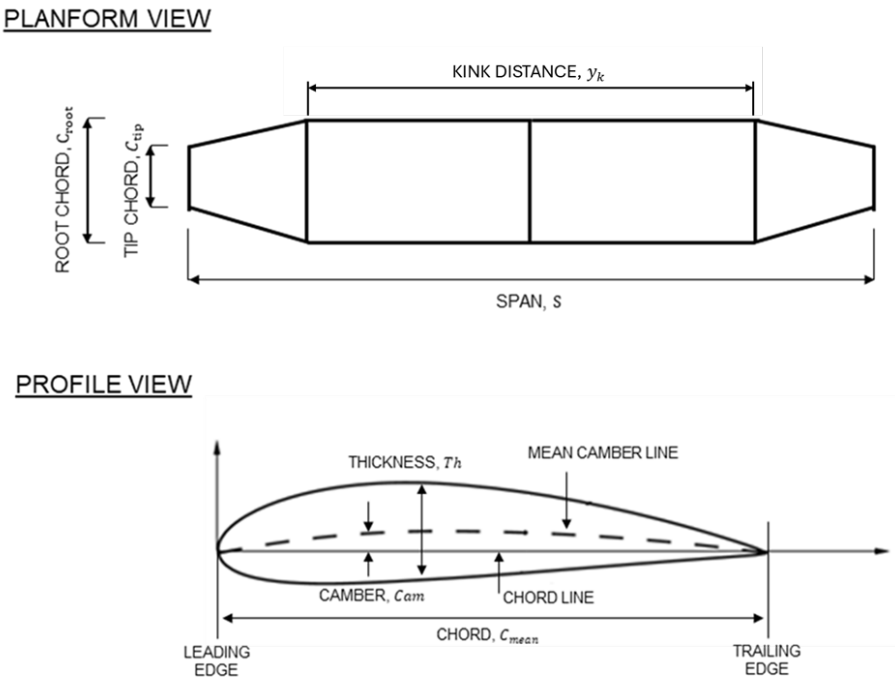


Figure 16: *Hydrofoil geometric definitions.*

A.1 Lift and Drag Coefficients

$$\text{Lift coefficient: } C_L(Re, \alpha, params) = \frac{L}{0.5\rho U^2 S}$$

$$\text{Drag coefficient: } C_D(Re, \alpha, params) = \frac{D}{0.5\rho U^2 S}$$



A.2 Equation of Motion

The second-order differential equation of motion for the body is given by:

$$A[\ddot{x}] + B[\dot{x}] + C[x] = F$$

where:

- A is the mass and added mass of the body in motion,
- B is the damping term,
- C is the restoring (spring) term,
- F is the sum of the excitation forces (including weight).

A.3 Reduced Frequency

The reduced frequency is defined as:

$$k = \frac{\omega b}{U}$$

where ω is the frequency of the incidence variations, b is the semi-chord of the hydrofoil, and U is the flow velocity.

A.4 Definitions of Parameters

- L and D: lift and drag forces, respectively,
- ρ: flow density,
- U: flow velocity,
- S: planform area (for a planar hydrofoil, this is the projected area of the foil in the direction of lift force at $\alpha = 0$),
- Re: Reynolds number,
- *Foil parameters*: geometric features such as aspect ratio, camber, and thickness,
- $[x]$: position of the body,
- $[\dot{x}]$: velocity (first derivative of position),



- $[\ddot{x}]$: acceleration (second derivative of position),
- h: draught of the trailing edge (bottom edge point of the foil),
- c: foil chord.

A.5 Dynamic Testing Procedure

The types of tests (summarised in Appendix II) aim to determine the coefficients and excitation forces (due to waves). The knowledge of these coefficients is crucial for modeling the dynamic behavior of the body in motion. Forces and moments acting on the foil are computed or measured, followed by harmonic analysis to evaluate the following:

- Coefficients A, B, and C for forced pitch and heave oscillations,
- Force F for incident wave conditions (foil advancing at a fixed angle of attack).

A.6 Test Conditions

The tests focused on hydrofoil forces (lift and drag) in both steady and unsteady conditions. The different types of runs carried out are as follows:

- (Quasi-)steady runs with the foil advancing at constant speed in calm water, varying immersion and/or angle of attack,
- Forced sinusoidal oscillations of pitch and heave while advancing in calm water, with multiple oscillation amplitudes and frequencies,
- Advancing in waves, characterized by varying combinations of amplitude and frequency, with the foil at a fixed angle of attack and speed.

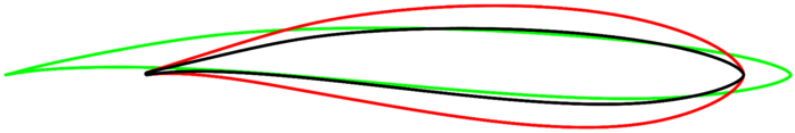
Appendix II: Model Test Campaign and Geometries

B Tested Foil Geometries

Three hydrofoil geometries were selected to represent the most relevant cases for experimental testing. The selection criteria included a simple planform shape composed of straight lines for ease of manufacturing at full scale. The geometries varied in aspect ratios (AR) and thickness ratios. Two aspect ratios, 4 and 8, were tested, with the higher aspect ratio (AR = 8) having two thickness ratios: 0.12 and 0.20. Figure 17 provides an overview of the foil identification names, profiles, and planform shapes.

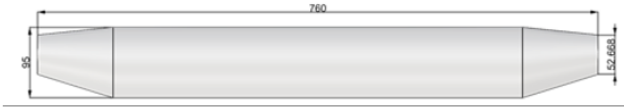


Foil profile based on Eppler 817:

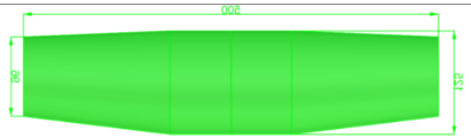


Foil ID	chord [m]	AR [-]	t/c [-]
I)	*0.095*	*8*	*0.12*
II)	*0.095*	*8*	*0.20*
III)	*0.095*	*4*	*0.20*

Planform Foil I and Foil II



Planform Foil III



foils as manufactured



Figure 17: *Foil geometries tested during the experimental campaign.*

B.1 Foil Selection and Design

The Eppler profile was chosen as the "optimal off-the-shelf" option for both fully foiling and foil-assisted modes due to its optimised shapes for low drag at both low and high speeds and its superior performance regarding cavitation resistance. This profile, combined with the simplified planform, served as the basis for exploring the design space and generating the training dataset for the tool. It also informed the design of cases for experimental validation.

B.2 Manufacture and Stiffness of Foils

The wings were manufactured through a collaborative effort between MARIN and a JIP participant. The objective was to produce simple, economical wings for experimental purposes. The wings were 3D-printed and reinforced with carbon rods to increase bending stiffness (as shown in Figure 16, "as manufactured"). Despite this effort, two of the foils (Foil I and Foil III), which had the smallest thickness ratios, exhibited noticeable elastic behavior during forced oscillations (pitch and heave) under both

quasi-steady and unsteady conditions.

From video observations (examples shown in Figure 18), Foil I and Foil III were deemed too flexible, and further analysis focused on Foil II, which demonstrated adequate rigidity. Foil II was thus selected as the primary reference for high-fidelity data acquisition.

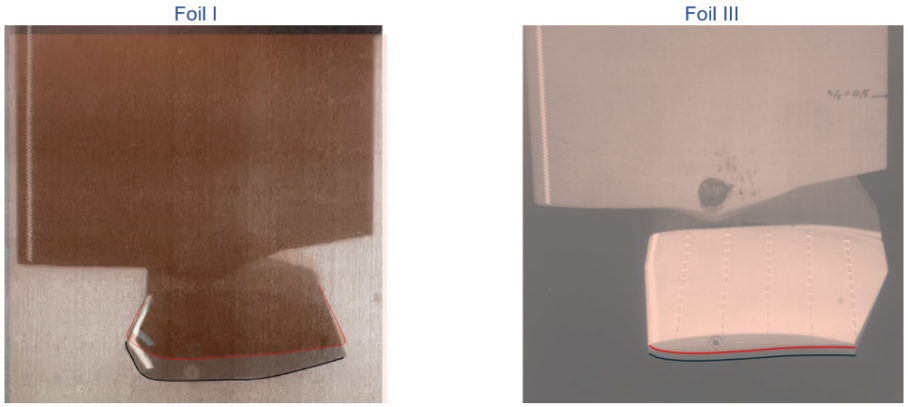


Figure 18: *Comparison of deformation (bending) in Foil I (left) and Foil III (right) during steady runs. The overlap shows the foil before (black) and after (red) acceleration. Clear bending of the wing tips can be seen. Videos (omitted here) emphasised this behavior during forced pitch and heave oscillations.*

B.3 Test Setup and Procedure

For the model tests conducted as part of the JIP, a custom test setup was developed, shown in Figures 19 and 20. A key feature of the setup was its ability to decouple pitch motion of the horizontal wing from the heave motion of the entire T-foil. The pitch motion was controlled by a new actuation system designed by one of the participants, Flying Fish. Heave motion was controlled by a hexapod beneath the pitch actuation system, which was mounted on a six-component force measurement frame.

The vertical strut was designed to minimise its impact on lift and drag, allowing for accurate comparisons of quasi-steady and unsteady conditions with an isolated horizontal wing.



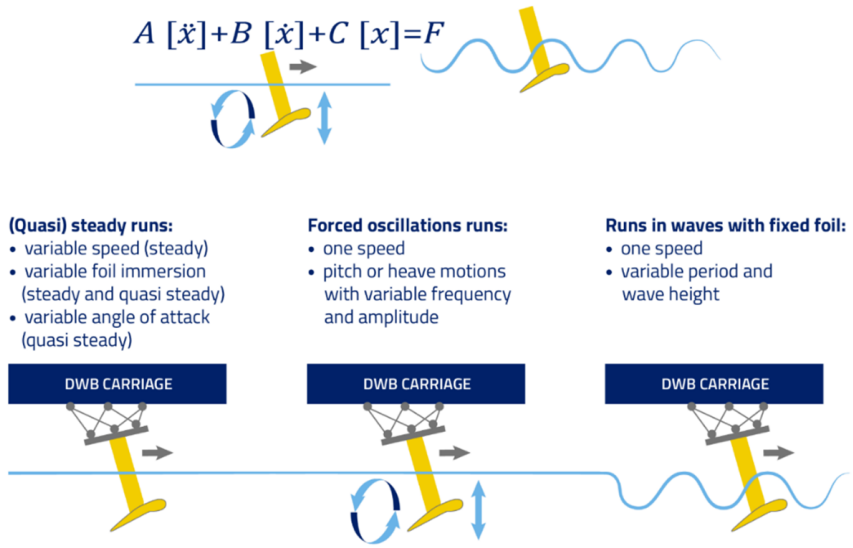


Figure 19: *Types of runs performed during the model test campaign (also replicated in CFD calculations) and their relationship with determining the coefficients of the equation of motion.*

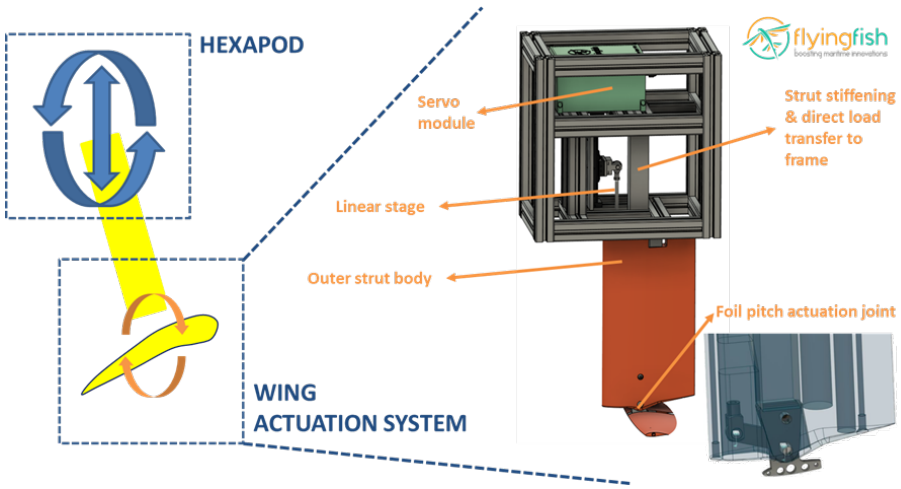


Figure 20: *Overview of the model test setup at TU Delft.*



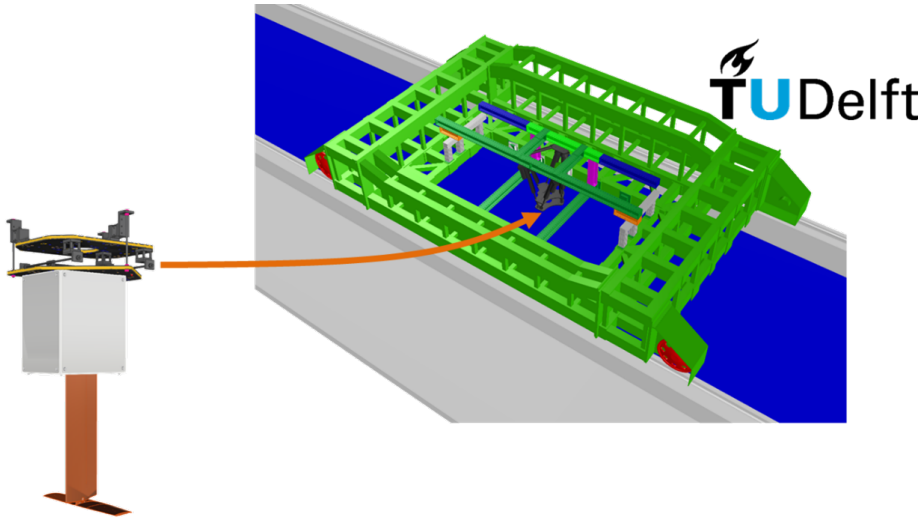


Figure 21: *Carriage at the TU Delft towing tank, showing the hexapod and assembly with a six-component frame for force measurements atop the pitch actuation system.*

Further details about the testing facility at TU Delft can be found at the following Link. The data from these tests provide critical insights into the dynamic behavior of the hydrofoils, particularly in terms of determining the coefficients of the equation of motion under various conditions, as summarised in the figures above.